

1. Administrative & Study Identification

Full Study Title A Study of Suvorexant (MK-4305) for the Treatment of Insomnia Disorder in Participants With Opioid Use Disorder (MK-4305-098) **Short Title/Acronym** Suvorexant for Insomnia in OUD **Protocol Number** MK-4305-098 **NCT Number** NCT06655883 **Sponsor Name** Merck **Investigational Product** Suvorexant (MK-4305 / Belsomra®) **ATC Code** N05CM19 **Phase of Development** Phase III **Trial Status** Recruiting **Medical Monitor** Clinical Director, Merck

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To evaluate whether suvorexant helps individuals with Opioid Use Disorder (OUD) fall asleep and sleep longer compared to placebo, by assessing the change in total sleep time. **Secondary Objectives:** To evaluate the safety, tolerability, and any adverse events associated with suvorexant administration in participants with OUD, as well as change in wakefulness after persistent sleep onset and substance abuse status.

2.2 Background & Rationale People with opioid use disorder (OUD) frequently experience significant trouble falling or staying asleep. Sleep disturbances can severely impact physical and psychological health and can increase the risk of relapse. Suvorexant is a dual orexin receptor antagonist already FDA-approved for treating insomnia. This study aims to evaluate if suvorexant is a safe and effective sleep aid for this specific population by reducing insomnia without interfering with their stable medications for opioid use disorder (MOUD).

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Randomized **Disease Areas:** Insomnia

3.2 Planned Enrollment & Duration **Target \$N\$:** 300 participants **Number of Sites:** Multicenter (including sites like Johns Hopkins University and Medical University of South Carolina) **Estimated Study Start:** 09-Oct-2025 **Estimated Primary Completion:** 24-Jun-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Has a primary diagnosis of Opioid Use Disorder (OUD) according to Diagnostic and Statistical Manual of Mental Disorder, 5th Edition (DSM-5), and confirmed through the Mini International Neuropsychiatric Interview (MINI). 2. Is on a verified, stable dose of medications for opioid use disorder (MOUD) treatment. 3. Meets DSM-5 criteria for the diagnosis of Insomnia Disorder. 4. Has a regular bedtime between 8 PM (20:00) and 1 AM (01:00) and is willing to maintain it for the duration of the study. 5. Has not used opioids for a period of at least 4 weeks before entering the study.

4.2 Exclusion Criteria 1. Has current uncontrolled major co-morbid psychiatric illness including major depressive disorder, bipolar disorder, schizophrenia, or any psychiatric condition with psychotic features. 2. Has current diagnosis or history within 5 years of any of the following: narcolepsy, sleep paralysis, severe periodic limb movement disorder, restless leg syndrome, cataplexy, circadian rhythm sleep disorder, parasomnia including nightmare disorder, sleep terror disorder, sleepwalking disorder, rapid eye movement (REM) behavior disorder, significant degree of sleep-related breathing disorder, excessive daytime sleepiness (EDS), or primary hypersomnia. 3. Is at imminent risk of self-harm. 4. Has a known history of stroke that may confound the diagnosis of insomnia. 5. Has a clinically significant movement disorder such as akinesia. 6. Has a history of hepatitis or liver disease. 7. Has habitual use of central nervous system (CNS)-depressants or stimulants that may be responsible for the participant's disturbed sleep. 8. Has a history of malignancy, ≤ 3 years prior to start of study, with the exception of nonmelanoma skin cancer, prostate cancer or localized carcinoma in situ of the cervix. 9. Has a history of hypersensitivity to more than 3 chemical classes of drugs, including prescription and over-the-counter medications. 10. Has donated blood products or had phlebotomy within 8 weeks prior to start of study. 11. Has a history of transmeridian travel within 2 weeks prior to start of study.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Participants receive 10 mg of suvorexant for the first three nights, increased to 20 mg of suvorexant nightly thereafter at the investigator's discretion. **Route of Administration:** Oral **Duration of Treatment:** Up to approximately 8 weeks.

5.2 Concomitant & Prohibited Therapy Permitted: Verified, stable doses of medications for opioid use disorder (MOUD). **Prohibited:** Use of schedule IV sleep medications, benzodiazepines, tranquilizers, or any medication that is strictly contraindicated with suvorexant.

6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :---
| :--- | :--- | | **Screening** | Day -14 to 0 | Informed Consent, Medical/Psychiatric History (MINI), Bedtime Verification | |
Baseline | Day 0 | Randomization, First Dose, Habitual Bedtime Polysomnography | | **Interim** | Week 4 \$\pm\$ 3 | Safety Labs, Efficacy Assessment, Dose Adjustment (if applicable) | | **End of Study** | Week 8 | Final Polysomnography, Vital Signs, Exit Interview, IP Accountability |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints 1. **Change from Baseline in Total Sleep Time at Week 8:** Measured in a sleep laboratory by polysomnography at the participant's habitual bedtime. 2. **Number of Participants Who Experience One or More Adverse Events (AEs):** An AE is any untoward medical occurrence in a clinical study participant, temporally associated with the use of study intervention, whether or not considered related to the study intervention. 3. **Number of Participants Who Experience One or More Serious Adverse Events (SAEs):** An SAE is any untoward medical occurrence that at any dose results in death, is life threatening, requires or prolongs inpatient hospitalization, results in persistent or significant disability or incapacity, is a congenital anomaly or birth defect or is another important medical event deemed such by medical or scientific judgement.

7.2 Secondary Endpoints 1. Change from Baseline in Wakefulness after Persistent Sleep Onset at Week 8. 2. Number of Participants Who are Positive for Substance Abuse at Day 1. 3. Number of Participants Who are Positive for Substance Abuse at Day 4.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous evaluation of untoward medical occurrences. **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Internal/External safety monitoring boards managed by Merck.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy evaluation (change in total sleep time), Safety Set for AE evaluations. **Sample Size Justification:** \$N = 300\$ participants powered to detect a statistically and clinically significant difference in total sleep time between the suvorexant and placebo

arms ($\alpha = 0.05$, $\text{Power} = 80\%$). **Handling of Missing Data:** Mixed-effect Model for Repeated Measures (MMRM) or Multiple Imputation as outlined in the SAP.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Routine onsite and remote monitoring visits to ensure data integrity and protocol adherence. **Audit Trail:** Robust tracking of eCRF locking and data clarification forms to ensure compliance with polysomnography and drug administration protocols.

11. Study Identifiers & Governance

Primary ID: MK-4305-098 **Registry Number:** NCT06655883 **Sponsor/Lead Organization:** Merck **Study Title:** A Study of Suvorexant (MK-4305) for the Treatment of Insomnia Disorder in Participants With Opioid Use Disorder (MK-4305-098) **Official Regulatory Title:** A Phase 3, Multicenter, Randomized, Double-blind, Placebo-controlled Study to Evaluate the Efficacy and Safety of Suvorexant for the Treatment of Insomnia in Participants With Opioid Use Disorder

12. Clinical Framework (Insomnia/ODU Specific)

Primary Condition: Insomnia Disorder in patients with Opioid Use Disorder (ODU) **Diagnosis Threshold:** DSM-5 criteria for Insomnia Disorder & ODU (confirmed via MINI) **Medical Methodology:** Dual orexin receptor antagonist therapy **Optimization Target:** Improvement in total sleep time and sleep maintenance

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for ODU incorporating monitored sleep therapy **Education Protocol (HCP):** Training on polysomnography procedures and suvorexant dose titration (10mg to 20mg) **Education Protocol (Patient):** Sleep hygiene counseling and adherence to strict regular bedtime routines (between 8 PM and 1 AM) **Quality Audit Mechanism:** Habitual bedtime adherence tracking and polysomnography data verification

14. Observation & Follow-Up Schedule

Total Duration: Approximately 8 weeks of treatment **Onsite Visit Cadence:** Baseline and Week 8 for overnight sleep laboratory assessments **Remote Monitoring Frequency:** Regular interval symptom and AE check-ins **Checklist Review Interval:** Sleep schedule tracking reviewed at each interim visit

15. Sign-off & Approval

Role	Name	Signature	Date	---	---	---	---
Principal Investigator	[Insert Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Insert Name]	_____	[DD-MMM-YYYY]				
Lead Statistician	[Insert Name]	_____	[DD-MMM-YYYY]				